

Tutorial: analysis of a research survey paper

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According to [ACM Computing Surveys](#), “a survey article assumes a general knowledge of the area; it emphasizes the classification of the existing literature, developing a perspective on the area, and evaluating trends.”

I. Objectives

The main objective of this tutorial (TD) is to read and analyze a scientific survey paper.

II. Introduction to efficient scientific paper reading

S. Keshav [1] proposes a three-pass method for efficiently reading a scientific paper. You will apply this method to reading the provided survey. Information below are directly extracted from [1]. During this tutorial, you will apply this reading method.

First pass: this is a quick scan of the paper

Expected duration: 10 minutes

Actions:

- “1. Carefully read the title, abstract, and introduction
2. Read the section and sub-section headings, but ignore everything else
3. Read the conclusions
4. Glance over the references, mentally ticking off the ones you’ve already read”

Available results/information after applying the first pass:

“1. Category: What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?

=> In our case, it is a survey paper.

2. Context: Which other papers is it related to? Which theoretical bases were used to analyze the problem?
3. Correctness: Do the assumptions appear to be valid?
4. Contributions: What are the paper’s main contributions?
5. Clarity: Is the paper well written?”

Second pass: careful reading of the paper and note-taking

Expected duration: one to two hours

Actions:

“1. Look carefully at the figures, diagrams and other illustrations in the paper. Pay special attention to graphs. Are the axes properly labeled? Are results shown with error bars, so that conclusions are statistically significant? Common mistakes like these will separate rushed, shoddy work from the truly excellent.

2. Remember to mark relevant unread references for further reading (this is a good way to learn more about the background of the paper).”

Available results/information after applying the first pass:

1. *You have understood the main content of the paper and you should be able to summarize it to someone else.*

At this point, you could miss some technical details from the paper or realize to you need further reading to understand the paper. You can take such actions if needed before resuming to the next pass of the proposed reading method.

Third pass: full understanding of the paper content to clearly identify the paper's contributions

Expected duration: up to five hours

Actions:

1. Give great attention to details, including proofs and technical aspects.
2. Challenge and criticize the provided assumptions, methods, results, tools...

Available results/information after applying the first pass:

1. You are able to identify strengths and weaknesses of the paper, including potential issues with the assumptions and/or the technical aspects.

III. Application to the provided survey paper and assignment

1. Read the provided survey paper, applying the above reading method.
2. Provide a detailed analysis of the structure of the paper.
3. Provide a detailed analysis of the introduction and the conclusion of the paper.
4. The survey paper provides a classification of existing works according to chosen criteria. Identify and analyze the classification criteria and the provided classes. Analyze the provided tables and figures if any.
5. Analyze the used references (date, journals/conference, authors, completeness, etc.),
6. Analyze the choice of the paper's title and sections headings, figures or tables captions, and explanations of figures and tables in the text.
7. Provide a (critical) presentation of 10 minutes to the whole class.

Reference:

[1] S. Keshav. “How to Read a Paper”, available at
<https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf>